
Engineering Mathematics – 1

Tutorial Solutions

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Note:

This document contains Tutorials 1 and 2 with short answers, followed by detailed solutions to selected problems. Try solving the questions yourself before looking at the solutions, as they are most useful for checking your work and understanding.

The notes are updated regularly, and any feedback, corrections, or suggestions for improvement are always welcome. If you spot an issue, please mention the page and relevant section, problem, figure, or equation number to help locate it quickly.

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Tutorial 1: Rolle's Theorem and Lagrange's Mean Value Theorem

Questions marked ► [Problem](#) are solved in full in Sections 1–2. Click the link to go to the solution.

1. Verify Rolle's theorem for the following functions.

(a) $f(x) = x^2(1-x)^2$ on $[0, 1]$. ► [Problem 1.1](#) [Ans. $c = \frac{1}{2}$]

(b) $f(x) = \frac{\sin x}{e^x}$ on $[0, \pi]$. [Ans. $c = \frac{\pi}{4}$]

(c) $f(x) = \tan x$ on $[0, \pi]$. [Ans. not applicable: $\tan x$ is discontinuous at $x = \frac{\pi}{2}$]

(d) $f(x) = e^x \sin x$ on $[0, \pi]$. ► [Problem 1.2](#) [Ans. $c = \frac{3\pi}{4}$]

(e) $f(x) = \log \left(\frac{x^2 + 9}{10x} \right)$ on $[1, 9]$. ► [Problem 1.3](#) [Ans. $c = 3$]

(f) $f(x) = \sqrt{(x-4)(5-x)}$ on $[4, 5]$. [Ans. $c = \frac{9}{2}$]

(g) $f(x) = \sqrt[3]{(x-2)^2}$ on $[1, 3]$. ► [Problem 1.4](#)
[Ans. not applicable: not differentiable at $x = 2$]

2. Apply Rolle's theorem to the function $f(x) = \sin x \sqrt{\cos x}$ on $[0, \frac{\pi}{2}]$, and hence show that $0 \leq f(x) \leq \sqrt[4]{\frac{4}{27}}$ for $x \in [0, \frac{\pi}{2}]$. [Ans. $c = \tan^{-1} \sqrt{2} \approx 0.9553$, $f(c) = \sqrt[4]{4/27} \approx 0.6204$]

3. Verify Lagrange's mean value theorem for the following functions.

(a) $f(x) = \cos^2 x$ on $[0, \frac{\pi}{2}]$. ► [Problem 2.1](#)
Ans. $c = \frac{1}{2} \sin^{-1} \left(\frac{2}{\pi} \right) \approx 0.3451$ or $c = \frac{1}{2} \left[\pi - \sin^{-1} \left(\frac{2}{\pi} \right) \right] \approx 1.2257$

(b) $f(x) = \sin^{-1} x$ on $[0, 1]$. ► [Problem 2.2](#) [Ans. $c = \sqrt{1 - \frac{4}{\pi^2}} \approx 0.7712$]

(c) $f(x) = 2x^3 - 3x + 1$ on $[-2, 2]$. ► [Problem 2.3](#) [Ans. $c = \pm \frac{2}{\sqrt{3}}$]

(d) $f(x) = (x-4)(x-6)(x-8)$ on $[4, 10]$. [Ans. $c = 8$]

(e) $f(x) = \arcsin x$ on $[0, 1]$ (the same function as (b)). [Ans. $c = \frac{\sqrt{\pi^2 - 4}}{\pi} \approx 0.7712$]

(f) $f(x) = \log x$ on $[1, e]$. [Ans. $c = e - 1$]

4. Apply Lagrange's mean value theorem to $f(x) = \log x$ to prove that $0 < \frac{1}{\log(1+x)} - \frac{1}{x} < 1$ for $x > 0$. ► [Problem 2.4](#)

5. For all $x > 0$, apply Lagrange's mean value theorem to prove the inequalities

$$(i) \cos x > 1 - \frac{x^2}{2}, \quad (ii) e^{-x} > 1 - x.$$

6. A ball is thrown vertically upward from the ground with an initial speed of 29.4 m/s. Its height above the ground after t seconds is $h(t) = 29.4t - 4.9t^2$ metres. Show that at some instant during the flight the ball is momentarily at rest, and find that instant and the maximum height attained. ► [Problem 1.5](#) [Ans. $c = 3$ s, $h(3) = 44.1$ m]

7. A car enters a toll booth at 8:00 a.m. and exits another booth 180 km away at 10:00 a.m. The entire highway has a posted speed limit of 70 km/h. On arrival the driver is issued a speeding citation based solely on the two timestamps, and disputes it, claiming that the speed never exceeded 70 km/h. Using the mean value theorem, justify the claim of the traffic authorities that the driver was speeding.

[Ans. $v(c) = 90$ km/h at some instant c , and $90 > 70$]

8. In an automated chemical batch reactor, an exothermic reaction is monitored over an 8-minute cycle, $t \in [0, 8]$. The temperature, in $^{\circ}\text{C}$, is $T(t) = 50 + 12t - 1.5t^2$. (a) Verify that T satisfies all the conditions of Rolle's theorem on $[0, 8]$. (b) Determine the critical time $c \in (0, 8)$ at which the instantaneous rate of temperature change vanishes, and the maximum temperature. ► Problem 1.6

[Ans. $T(0) = T(8) = 50^{\circ}\text{C}$, $c = 4$ min, $T(4) = 74^{\circ}\text{C}$]

9. A storage tank holds 200 litres of liquid at 9:00 a.m. The inlet and outlet valves are rated so that the volume in the tank can never change at a rate exceeding 12 L/min in either direction. An operator's log records that the tank held 500 litres at 9:20 a.m. Show that this entry must be erroneous.

Ans. $|V(20) - 200| \leq 12 \times 20 = 240$, so $V(20) \leq 440 < 500$

10. A simply supported beam of span $L = 6$ m carries a load varying linearly from zero at the left support A to a maximum intensity w_0 at the right support B . Measuring x from A , the deflection is

$$y(x) = Kx(x^4 - 120x^2 + 3024), \quad K = \frac{w_0}{720EI} > 0.$$

Show that a point of maximum deflection exists strictly inside the span, and locate it.

Ans. $c \approx 3.116 \text{ m} = 0.5193L$, $y(c) \approx 0.00652 \frac{w_0 L^4}{EI}$

11. A simplified cam gives the inlet valve lift over a 240° crank-angle event as $L(\theta) = \frac{\theta(240 - \theta)^2}{204800}$ mm, $0 \leq \theta \leq 240$. Show that the valve attains a maximum lift at which it is momentarily stationary, and determine the maximum lift and the corresponding crank angle.

[Ans. $\theta = 80^{\circ}$, $L(80) = 10$ mm]

12. Suppose that f is continuous on $[-7, 0]$ and differentiable on $(-7, 0)$, with $f(-7) = -3$ and $f'(x) \leq 2$ for all $x \in (-7, 0)$. Determine the largest possible value of $f(0)$. ► Problem 2.5

[Ans. 11]

Tutorial 2: Series and Taylor–Maclaurin Series

Questions marked ► [Problem](#) are solved in full in Sections 3–4. Click the link to go to the solution.

Series

1. Test the convergence of the following series by the root test.

(a) $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2} \right)^n$. ► [Problem 3.1](#) [**Ans.** $L = \frac{2}{3} < 1$: convergent]

(b) $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n} \right)^{-n^2}$. ► [Problem 3.2](#) [**Ans.** $L = \frac{1}{e} < 1$: convergent]

(c) $\left(\frac{2^2}{1^2} - \frac{2}{1} \right)^{-1} + \left(\frac{3^3}{2^3} - \frac{3}{2} \right)^{-2} + \left(\frac{4^4}{3^4} - \frac{4}{3} \right)^{-3} + \dots$ [**Ans.** $L = \frac{1}{e-1} < 1$: convergent]

(d) $\sum_{n=1}^{\infty} \frac{7}{(2n+5)^n}$. [**Ans.** $L = 0 < 1$: convergent]

(e) $\sum_{n=1}^{\infty} \left(\frac{4n+3}{3n-5} \right)^n$. [**Ans.** $L = \frac{4}{3} > 1$: divergent]

(f) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$. [**Ans.** $L = \frac{1}{2} < 1$: convergent]

(g) $\sum_{n=1}^{\infty} \frac{4^n}{(3n)^n}$. [**Ans.** $L = 0 < 1$: convergent]

(h) $\sum_{n=1}^{\infty} \left(1 + \frac{2}{n} \right)^{n^2}$. ► [Problem 3.3](#) [**Ans.** $L = e^2 > 1$: divergent]

(i) $\sum_{n=1}^{\infty} \frac{1}{2^n} \left(1 + \frac{1}{n} \right)^{n^2}$. [**Ans.** $L = \frac{e}{2} > 1$: divergent]

(j) $\sum_{n=2}^{\infty} (\log n)^{-2n}$. [**Ans.** $L = 0 < 1$: convergent]

(k) $\sum_{n=1}^{\infty} \frac{1}{\left(1 + \frac{1}{\sqrt{n}} \right)^{n^{3/2}}}$. [**Ans.** $L = \frac{1}{e} < 1$: convergent]

2. Test the convergence of the following series by the ratio test.

(a) $\sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n + 1}$. ► [Problem 3.4](#) [**Ans.** $L = \frac{2}{3} < 1$: convergent]

(b) $\sum_{n=1}^{\infty} \frac{n! 3^n}{n^n}$. [**Ans.** $L = \frac{3}{e} > 1$: divergent]

(c) $\sum_{n=0}^{\infty} \frac{2^n + 5}{3^n}$. [**Ans.** $L = \frac{2}{3} < 1$: convergent]

(d) $\sum_{n=0}^{\infty} \frac{n^3 + a}{2^n + a}$, where $a > 0$. ▶ Problem 3.5 [Ans. $L = \frac{1}{2} < 1$: convergent]

(e) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$. [Ans. $L = \frac{1}{2} < 1$: convergent]

(f) $\sum_{n=2}^{\infty} \frac{3^{n+2}}{\log n}$. ▶ Problem 3.6 [Ans. $L = 3 > 1$: divergent]

(g) $\sum_{n=1}^{\infty} \frac{n^p}{n!}$. [Ans. $L = 0 < 1$: convergent for every real p]

(h) $\sum_{n=1}^{\infty} \frac{n!}{4^{n-3}}$. [Ans. $L = \infty$: divergent]

(i) $\sum_{n=1}^{\infty} \frac{8n - 4}{9n + 9}$.

Ans. $L = 1$, so the ratio test is inconclusive. Since $a_n \rightarrow \frac{8}{9} \neq 0$, the series diverges by the n th term test.

3. Examine the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{nx}{n+1} \right)^n$, where x is real. ▶ Problem 3.7
[Ans. convergent for $|x| < 1$, divergent for $|x| \geq 1$]

4. Discuss the convergence of the series $1 + \frac{2!}{2^2} + \frac{3!}{3^3} + \frac{4!}{4^4} + \dots$.
[Ans. $a_n = \frac{n!}{n^n}$, $L = \frac{1}{e} < 1$: convergent]

5. Discuss the convergence of the series $\frac{x}{1+x} + \frac{x^2}{1+x^2} + \frac{x^3}{1+x^3} + \dots + \frac{x^n}{1+x^n} + \dots$, where $x > 0$.
[Ans. convergent for $0 < x < 1$, divergent for $x \geq 1$]

Taylor and Maclaurin series

1. Obtain the Maclaurin series expansion, as an infinite series, of each of the following functions. ▶ Problem 4.1

(a) $\sin x$. [Ans. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$]

(b) $\cos x$. [Ans. $\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$]

(c) e^x . [Ans. $\sum_{n=0}^{\infty} \frac{x^n}{n!}$]

(d) $\log(1+x)$. [Ans. $\sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}$]

(e) $(1+x)^k$.

Ans. $1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \frac{k(k-1)(k-2)(k-3)}{4!}x^4 + \dots$

2. Obtain the Taylor series expansion of $\log_e x$ about $x = 1$ up to the term containing $(x - 1)^4$, and hence find the approximate value of $\log_e(1.1)$. ► Problem 4.2

Ans. $(x - 1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{3} - \frac{(x-1)^4}{4}$, $\log(1.1) \approx 0.0953$

3. Obtain the Taylor series expansion of $\log(\cos x)$ about $x = \frac{\pi}{3}$ up to the fourth-degree term.

Ans. $-\log 2 - \sqrt{3} \left(x - \frac{\pi}{3}\right) - 2 \left(x - \frac{\pi}{3}\right)^2 - \frac{4}{\sqrt{3}} \left(x - \frac{\pi}{3}\right)^3 - \frac{10}{3} \left(x - \frac{\pi}{3}\right)^4$

4. Obtain the Taylor series expansion of $f(x) = \sin x$ about $x = \frac{\pi}{6}$ up to the term containing $\left(x - \frac{\pi}{6}\right)^5$. ► Problem 4.3

Ans. $\frac{1}{2} + \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{6}\right) - \frac{1}{4} \left(x - \frac{\pi}{6}\right)^2 - \frac{\sqrt{3}}{12} \left(x - \frac{\pi}{6}\right)^3 + \frac{1}{48} \left(x - \frac{\pi}{6}\right)^4 + \frac{\sqrt{3}}{240} \left(x - \frac{\pi}{6}\right)^5$

5. Expand $\tan^{-1} x$ in powers of $(x - 1)$ up to the term containing $(x - 1)^4$.

Ans. $\frac{\pi}{4} + \frac{1}{2} \left[(x - 1) - \frac{(x-1)^2}{2} + \frac{(x-1)^3}{6} \right]$ (the $(x - 1)^4$ term is 0)

6. Expand $\log(\sec x)$ up to the term containing x^6 using Maclaurin's series.

[**Ans.** $\frac{x^2}{2} + \frac{x^4}{12} + \frac{x^6}{45}$]

7. Expand $e^{\sin x}$ by Maclaurin's series up to the term containing x^4 . [**Ans.** $1 + x + \frac{x^2}{2} - \frac{x^4}{8}$]

8. Obtain the Maclaurin expansion of a^x up to the third-degree term.

Ans. $1 + (\log a)x + \frac{(\log a)^2}{2}x^2 + \frac{(\log a)^3}{6}x^3$

9. Obtain $\log(1 + \cos x)$ by Maclaurin's series up to the term containing x^4 .

[**Ans.** $\log 2 - \frac{x^2}{4} - \frac{x^4}{96}$]

10. Expand $\tan x$ about $x = \frac{\pi}{4}$ up to the third-degree term, and hence find $\tan 46^\circ$.

Ans. $1 + 2 \left(x - \frac{\pi}{4}\right) + 2 \left(x - \frac{\pi}{4}\right)^2 + \frac{8}{3} \left(x - \frac{\pi}{4}\right)^3$, $\tan 46^\circ \approx 1.0355$

11. Evaluate $\int e^{-x^2} dx$ as an infinite series. ► Problem 4.4 [**Ans.** $x - \frac{x^3}{3} + \frac{x^5}{10} - \dots + C$]

1 Rolle's Theorem

Theorem 1.1: Rolle's theorem

Let f be a function that satisfies the following three conditions:

- (i) f is continuous on the closed interval $[a, b]$.
- (ii) f is differentiable on the open interval (a, b) .
- (iii) $f(a) = f(b)$.

Then there is a number $c \in (a, b)$ such that $f'(c) = 0$.

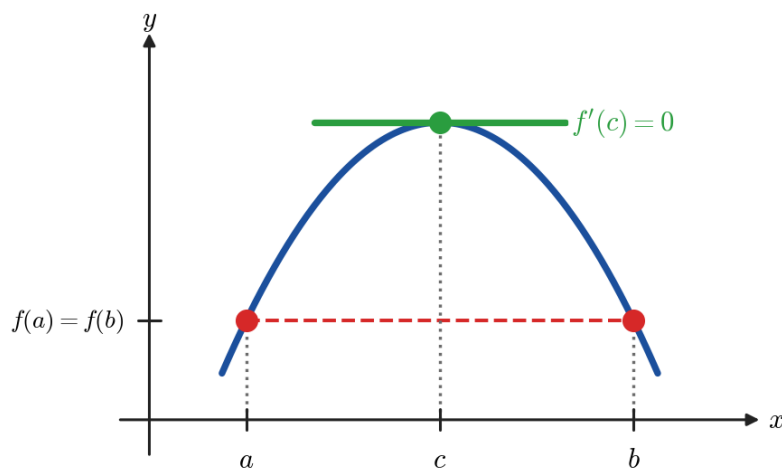


Figure 1.1: When the curve starts and ends at the same height, it must turn somewhere in between, and there the tangent is horizontal.

Note:

How to verify Rolle's theorem.

1. Write down the given data: $f(x)$, a and b .
2. Check conditions (i) and (ii), and evaluate $f(a)$ and $f(b)$ by substitution to check (iii).
3. Differentiate to get $f'(x)$, and write $f'(c)$.
4. Solve $f'(c) = 0$ and keep only the solutions that lie strictly inside (a, b) .

If a condition fails, the theorem does not apply — and then a point with $f'(c) = 0$ may or may not exist.

Note:

Sums, products, quotients (with non-zero denominator) and composites of polynomials, exponential, logarithmic and trigonometric functions are continuous and differentiable wherever they are defined. This settles conditions (i) and (ii) for most functions met in practice.

1.1 Worked problems

Problem 1.1. [Tutorial 1, Q1(a)] Verify Rolle's theorem for $f(x) = x^2(1-x)^2$ on $[0, 1]$.

Solution: We are given $f(x) = x^2(1-x)^2$, with $a = 0$ and $b = 1$.

We check the conditions of Rolle's theorem.

- (i) f is a polynomial, so it is continuous on $[0, 1]$.
- (ii) f is a polynomial, so it is differentiable on $(0, 1)$.
- (iii) Substituting the end points,

$$f(a) = f(0) = 0^2(1-0)^2 = 0 \cdot 1 = 0, \quad f(b) = f(1) = 1^2(1-1)^2 = 1 \cdot 0 = 0.$$

Hence $f(a) = f(b)$.

All three conditions of Rolle's theorem are satisfied. By Rolle's theorem, there exists a $c \in (0, 1)$ such that $f'(c) = 0$.

Next, we differentiate f . First multiply out, using $(1-x)^2 = 1 - 2x + x^2$:

$$f(x) = x^2(1 - 2x + x^2) = x^2 - 2x^3 + x^4.$$

Using $\frac{d}{dx}(x^n) = nx^{n-1}$ term by term,

$$f'(x) = 2x - 2(3x^2) + 4x^3 = 2x - 6x^2 + 4x^3, \quad \text{so} \quad f'(c) = 2c - 6c^2 + 4c^3.$$

To find c , we solve $f'(c) = 0$:

$$2c - 6c^2 + 4c^3 = 0.$$

Taking out the common factor $2c$, we get

$$2c(2c^2 - 3c + 1) = 0.$$

Factorising the quadratic $2c^2 - 3c + 1 = (2c - 1)(c - 1)$, we get

$$2c(2c - 1)(c - 1) = 0.$$

Setting each factor equal to zero, we get $c = 0$, $c = \frac{1}{2}$ or $c = 1$. The values $c = 0$ and $c = 1$ are end points and are not in $(0, 1)$. The value $c = \frac{1}{2}$ lies in $(0, 1)$.

Rolle's theorem is verified, with $c = \frac{1}{2} \in (0, 1)$.

Problem 1.2. [Tutorial 1, Q1(d)] Verify Rolle's theorem for $f(x) = e^x \sin x$ on $[0, \pi]$.

Solution: We are given $f(x) = e^x \sin x$, with $a = 0$ and $b = \pi$.

We check the conditions of Rolle's theorem.

- (i) e^x and $\sin x$ are continuous for all x , so their product is continuous on $[0, \pi]$.
- (ii) e^x and $\sin x$ are differentiable for all x , so their product is differentiable on $(0, \pi)$.
- (iii) Substituting the end points,

$$f(a) = f(0) = e^0 \sin 0 = 1 \cdot 0 = 0, \quad f(b) = f(\pi) = e^\pi \sin \pi = e^\pi \cdot 0 = 0.$$

Hence $f(a) = f(b)$.

All three conditions of Rolle's theorem are satisfied. By Rolle's theorem, there exists a $c \in (0, \pi)$ such that $f'(c) = 0$.

Next, we differentiate f . By the product rule $(uv)' = u'v + uv'$, with $u = e^x$, $u' = e^x$, $v = \sin x$, $v' = \cos x$:

$$f'(x) = e^x \sin x + e^x \cos x = e^x (\sin x + \cos x), \quad \text{so} \quad f'(c) = e^c (\sin c + \cos c).$$

To find c , we solve $f'(c) = 0$:

$$e^c (\sin c + \cos c) = 0.$$

Since $e^c > 0$ for every c , we must have

$$\sin c + \cos c = 0.$$

Moving $\cos c$ to the right-hand side, we get

$$\sin c = -\cos c.$$

Dividing both sides by $\cos c$ (which is non-zero here, since $\cos c = 0$ would force $\sin c = 0$ as well, which is impossible), we get

$$\frac{\sin c}{\cos c} = -1, \quad \text{that is,} \quad \tan c = -1.$$

In $(0, \pi)$ the tangent is negative only in the second quadrant. Since $\tan \frac{\pi}{4} = 1$ and $\tan(\pi - \theta) = -\tan \theta$,

$$c = \pi - \frac{\pi}{4} = \frac{3\pi}{4} \in (0, \pi).$$

Rolle's theorem is verified, with $c = \frac{3\pi}{4} \in (0, \pi)$.

Problem 1.3. [Tutorial 1, Q1(e)] Verify Rolle's theorem for $f(x) = \log \left(\frac{x^2 + 9}{10x} \right)$ on $[1, 9]$.

Solution: We are given $f(x) = \log \left(\frac{x^2 + 9}{10x} \right)$, with $a = 1$ and $b = 9$.

We check the conditions of Rolle's theorem.

- (i) For $x \in [1, 9]$, the quantity $\frac{x^2 + 9}{10x}$ is continuous and positive, so its logarithm is defined and continuous on $[1, 9]$.
- (ii) For the same reason, f is differentiable on $(1, 9)$.
- (iii) Substituting the end points,

$$f(a) = f(1) = \log \left(\frac{1^2 + 9}{10(1)} \right) = \log \frac{10}{10} = \log 1 = 0,$$

$$f(b) = f(9) = \log \left(\frac{9^2 + 9}{10(9)} \right) = \log \frac{81 + 9}{90} = \log \frac{90}{90} = \log 1 = 0.$$

Hence $f(a) = f(b)$.

All three conditions of Rolle's theorem are satisfied. By Rolle's theorem, there exists a $c \in (1, 9)$ such that $f'(c) = 0$.

Before differentiating, we simplify f . Using $\log \frac{A}{B} = \log A - \log B$ and $\log(AB) = \log A + \log B$,

$$f(x) = \log(x^2 + 9) - \log(10x) = \log(x^2 + 9) - \log 10 - \log x.$$

Next, we differentiate f . Using $\frac{d}{dx}[\log u] = \frac{u'}{u}$, $\frac{d}{dx}(\log x) = \frac{1}{x}$, and $\frac{d}{dx}(\log 10) = 0$ (a constant),

$$f'(x) = \frac{2x}{x^2 + 9} - 0 - \frac{1}{x}.$$

Taking the common denominator $x(x^2 + 9)$,

$$f'(x) = \frac{2x \cdot x - (x^2 + 9)}{x(x^2 + 9)} = \frac{2x^2 - x^2 - 9}{x(x^2 + 9)} = \frac{x^2 - 9}{x(x^2 + 9)}, \quad \text{so} \quad f'(c) = \frac{c^2 - 9}{c(c^2 + 9)}.$$

To find c , we solve $f'(c) = 0$. A fraction is zero only when its numerator is zero, so

$$c^2 - 9 = 0.$$

Moving 9 to the right-hand side, we get $c^2 = 9$. Taking the square root on both sides, we get

$$c = \pm 3.$$

Of these, $c = -3 \notin (1, 9)$ and $c = 3 \in (1, 9)$.

Rolle's theorem is verified, with $c = 3 \in (1, 9)$.

Problem 1.4. [Tutorial 1, Q1(g)] Verify Rolle's theorem for $f(x) = \sqrt[3]{(x-2)^2}$ on $[1, 3]$.

Solution: We are given $f(x) = \sqrt[3]{(x-2)^2} = (x-2)^{2/3}$, with $a = 1$ and $b = 3$.

We check the conditions of Rolle's theorem.

(i) f is the cube root of the polynomial $(x-2)^2$, so it is continuous on $[1, 3]$.

(ii) Using $\frac{d}{dx}[(x-2)^n] = n(x-2)^{n-1}$ with $n = \frac{2}{3}$,

$$f'(x) = \frac{2}{3}(x-2)^{\frac{2}{3}-1} = \frac{2}{3}(x-2)^{-1/3} = \frac{2}{3\sqrt[3]{x-2}}.$$

At $x = 2$ the denominator is $3\sqrt[3]{0} = 0$, so $f'(2)$ does not exist. Since $2 \in (1, 3)$, f is *not* differentiable on $(1, 3)$.

(iii) Substituting the end points,

$$f(a) = f(1) = \sqrt[3]{(1-2)^2} = \sqrt[3]{(-1)^2} = \sqrt[3]{1} = 1,$$

$$f(b) = f(3) = \sqrt[3]{(3-2)^2} = \sqrt[3]{1^2} = \sqrt[3]{1} = 1.$$

Hence $f(a) = f(b)$.

Condition (ii) fails, so Rolle's theorem cannot be applied. In fact the conclusion also fails: for $c \neq 2$,

$$f'(c) = \frac{2}{3\sqrt[3]{c-2}},$$

and $f'(c) = 0$ would require the numerator 2 to be 0, which is impossible. So there is no $c \in (1, 3)$ with $f'(c) = 0$.

Rolle's theorem is not applicable: f is not differentiable at $x = 2 \in (1, 3)$, and no c with $f'(c) = 0$ exists.

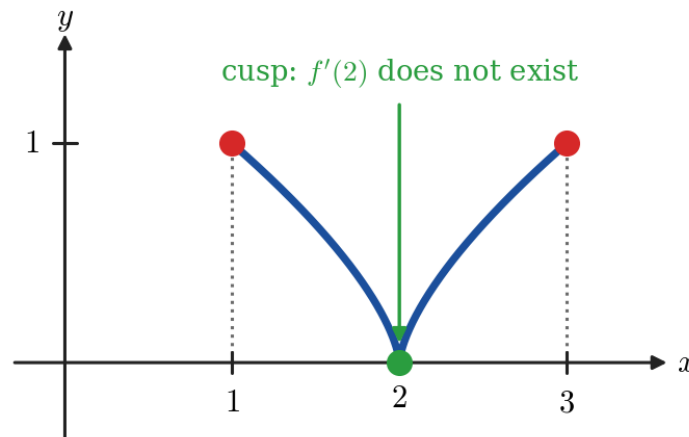


Figure 1.2: $f(x) = \sqrt[3]{(x-2)^2}$ on $[1, 3]$. The ends are at the same height, but the curve turns at a sharp point rather than smoothly, so there is no horizontal tangent.

Problem 1.5. [Tutorial 1, Q6] A ball is thrown vertically upward from the ground with an initial speed of 29.4 m/s. Its height above the ground after t seconds is

$$h(t) = 29.4t - 4.9t^2 \quad \text{metres.}$$

Show that at some instant during the flight the ball is momentarily at rest, and find that instant and the maximum height attained.

Solution: We are given $h(t) = 29.4t - 4.9t^2$.

The flight starts and ends when the ball is on the ground, that is, when $h(t) = 0$:

$$29.4t - 4.9t^2 = 0.$$

Taking out the common factor $4.9t$ (note that $29.4 = 4.9 \times 6$), we get

$$4.9t(6 - t) = 0.$$

Setting each factor equal to zero, we get $t = 0$ or $t = 6$. So $a = 0$ and $b = 6$.

We check the conditions of Rolle's theorem.

- (i) h is a polynomial, so it is continuous on $[0, 6]$.
- (ii) h is a polynomial, so it is differentiable on $(0, 6)$.
- (iii) From the above, we have $h(a) = h(0) = 0 = h(6) = h(b)$.

All three conditions of Rolle's theorem are satisfied. By Rolle's theorem, there exists a $c \in (0, 6)$ such that $h'(c) = 0$. Since $h'(t)$ is the velocity of the ball, the ball is momentarily at rest at $t = c$.

To find c , we differentiate h . Using $\frac{d}{dt}(t^n) = nt^{n-1}$,

$$h'(t) = 29.4 - 4.9(2t) = 29.4 - 9.8t, \quad \text{so} \quad h'(c) = 29.4 - 9.8c.$$

Solving $h'(c) = 0$:

$$29.4 - 9.8c = 0.$$

Moving $9.8c$ to the right-hand side, we get $9.8c = 29.4$. Dividing both sides by 9.8, we get

$$c = \frac{29.4}{9.8} = 3 \text{ s}, \quad 3 \in (0, 6).$$

For the maximum height, note that $h''(t) = -9.8 < 0$, so h has a maximum at $t = 3$:

$$h(3) = 29.4(3) - 4.9(3)^2 = 88.2 - 4.9(9) = 88.2 - 44.1 = 44.1 \text{ m}.$$

The ball is momentarily at rest at $t = 3$ s, and the maximum height is 44.1 m.

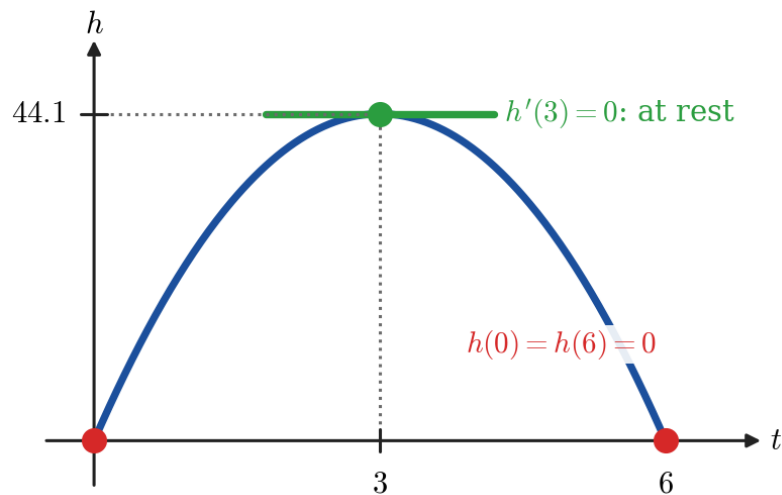


Figure 1.3: Height of the ball. It leaves and returns to the ground, so by Rolle's theorem its velocity $h'(t)$ is zero at some instant: the top of the flight, $t = 3$ s.

Problem 1.6. [Tutorial 1, Q8] In an automated chemical batch reactor, an exothermic reaction is monitored over an 8-minute cycle, $t \in [0, 8]$. The temperature $T(t)$, in $^{\circ}\text{C}$, is given by

$$T(t) = 50 + 12t - 1.5t^2.$$

- (a) Verify that T satisfies all the conditions of Rolle's theorem on $[0, 8]$.
- (b) Determine the critical time $c \in (0, 8)$ at which the instantaneous rate of temperature change vanishes, and find the maximum temperature.

Solution: We are given $T(t) = 50 + 12t - 1.5t^2$, with $a = 0$ and $b = 8$.

- (a) (i) T is a polynomial, so it is continuous on $[0, 8]$.

(ii) T is a polynomial, so it is differentiable on $(0, 8)$.

(iii) Substituting the end points,

$$T(a) = T(0) = 50 + 12(0) - 1.5(0)^2 = 50 + 0 - 0 = 50,$$

$$T(b) = T(8) = 50 + 12(8) - 1.5(8)^2 = 50 + 96 - 1.5(64) = 50 + 96 - 96 = 50.$$

$$\text{Hence } T(a) = T(b) = 50^\circ\text{C}.$$

All three conditions of Rolle's theorem are satisfied.

(b) Using $\frac{d}{dt}(t^n) = nt^{n-1}$, the rate of change of temperature is

$$T'(t) = 0 + 12 - 1.5(2t) = 12 - 3t, \quad \text{so} \quad T'(c) = 12 - 3c.$$

By Rolle's theorem, there exists a $c \in (0, 8)$ such that $T'(c) = 0$. Solving,

$$12 - 3c = 0.$$

Moving $3c$ to the right-hand side, we get $3c = 12$. Dividing both sides by 3, we get

$$c = 4 \text{ minutes}, \quad 4 \in (0, 8).$$

Since $T''(t) = -3 < 0$, T has a maximum at $t = 4$:

$$T(4) = 50 + 12(4) - 1.5(4)^2 = 50 + 48 - 1.5(16) = 50 + 48 - 24 = 74^\circ\text{C}.$$

(a) All three conditions hold, with $T(0) = T(8) = 50^\circ\text{C}$. (b) $c = 4$ minutes, and the maximum temperature is 74°C .

2 Lagrange's Mean Value Theorem

Theorem 2.1: Lagrange's mean value theorem

Let f be a function that satisfies the following conditions:

- (i) f is continuous on the closed interval $[a, b]$.
- (ii) f is differentiable on the open interval (a, b) .

Then there is a number $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}. \quad (2.1)$$

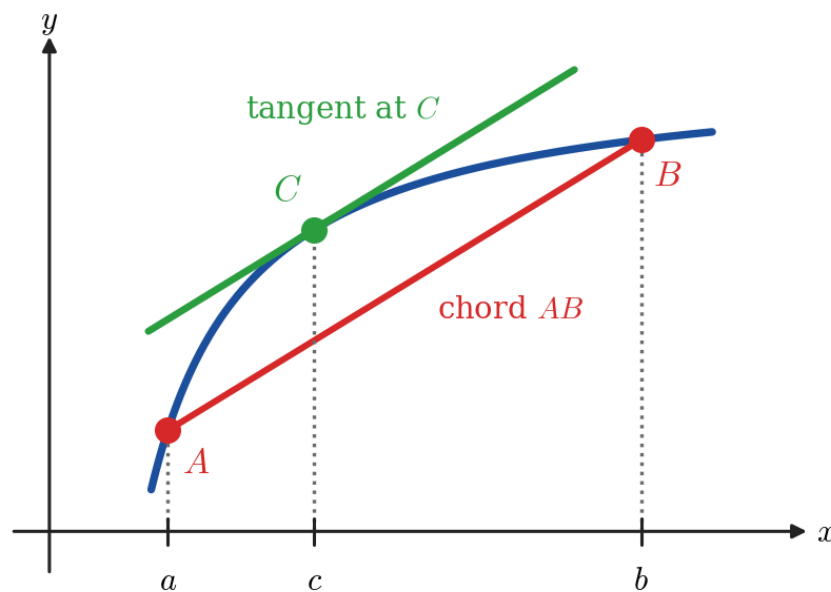


Figure 2.1: At some point C between A and B the tangent is parallel to the chord AB : its slope $f'(c)$ equals the slope $\frac{f(b)-f(a)}{b-a}$ of the chord.

Note:

How to verify Lagrange's theorem.

1. Write down the given data: $f(x)$, a and b .
2. Check conditions (i) and (ii).
3. Evaluate $f(a)$ and $f(b)$ by substitution, and compute the slope $\frac{f(b)-f(a)}{b-a}$ of the chord.
4. Differentiate to get $f'(x)$, write $f'(c)$, and solve equation (2.1), keeping only solutions in (a, b) .

Note:

Multiplying (2.1) by $b - a$ gives $f(b) - f(a) = (b - a)f'(c)$. This turns information about the derivative into information about the function: for instance, if $f'(x) \leq M$ throughout (a, b) , then $f(b) - f(a) \leq M(b - a)$. This is how the theorem proves inequalities and bounds.

2.1 Worked problems

Problem 2.1. [Tutorial 1, Q3(a)] Verify Lagrange's mean value theorem for $f(x) = \cos^2 x$ on $[0, \frac{\pi}{2}]$.

Solution: We are given $f(x) = \cos^2 x$, with $a = 0$ and $b = \frac{\pi}{2}$.

We check the conditions of Lagrange's mean value theorem.

(i) $\cos x$ is continuous for all x , so $\cos^2 x$ is continuous on $[0, \frac{\pi}{2}]$.

(ii) $\cos x$ is differentiable for all x , so $\cos^2 x$ is differentiable on $(0, \frac{\pi}{2})$.

Both conditions are satisfied.

Next, we compute the slope of the chord. Substituting the end points,

$$f(a) = f(0) = \cos^2 0 = (1)^2 = 1, \quad f(b) = f\left(\frac{\pi}{2}\right) = \cos^2 \frac{\pi}{2} = (0)^2 = 0,$$

so

$$\frac{f(b) - f(a)}{b - a} = \frac{0 - 1}{\frac{\pi}{2} - 0} = \frac{-1}{\pi/2} = -\frac{2}{\pi}.$$

Next, we differentiate f . By the chain rule,

$$\frac{d}{dx}[(\cos x)^2] = 2 \cos x \cdot \frac{d}{dx}(\cos x),$$

and $\frac{d}{dx}(\cos x) = -\sin x$. Using $2 \sin x \cos x = \sin 2x$,

$$f'(x) = 2 \cos x (-\sin x) = -2 \sin x \cos x = -\sin 2x, \quad \text{so} \quad f'(c) = -\sin 2c.$$

By LMVT, there exists a $c \in (0, \frac{\pi}{2})$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$. That is,

$$-\sin 2c = -\frac{2}{\pi}.$$

Multiplying both sides by -1 , we get

$$\sin 2c = \frac{2}{\pi} \approx 0.6366.$$

For $c \in (0, \frac{\pi}{2})$ we have $2c \in (0, \pi)$. In $(0, \pi)$ the equation $\sin \theta = k$ with $0 < k < 1$ has two solutions, $\theta = \sin^{-1} k$ and $\theta = \pi - \sin^{-1} k$. Hence

$$2c = \sin^{-1}\left(\frac{2}{\pi}\right) \quad \text{or} \quad 2c = \pi - \sin^{-1}\left(\frac{2}{\pi}\right),$$

$$c = \frac{1}{2} \sin^{-1}\left(\frac{2}{\pi}\right) \approx \frac{0.6901}{2} \approx 0.3451 \quad \text{or} \quad c = \frac{1}{2} \left[\pi - \sin^{-1}\left(\frac{2}{\pi}\right) \right] \approx \frac{2.4515}{2} \approx 1.2257.$$

Both values lie in $(0, \frac{\pi}{2}) \approx (0, 1.5708)$.

Lagrange's mean value theorem is verified, with $c = \frac{1}{2} \sin^{-1}\left(\frac{2}{\pi}\right) \approx 0.3451$ or $c = \frac{1}{2} \left[\pi - \sin^{-1}\left(\frac{2}{\pi}\right) \right] \approx 1.2257$.

Problem 2.2. [Tutorial 1, Q3(b)] Verify Lagrange's mean value theorem for $f(x) = \sin^{-1} x$ on $[0, 1]$.

Solution: We are given $f(x) = \sin^{-1} x$, with $a = 0$ and $b = 1$.

We check the conditions of Lagrange's mean value theorem.

- (i) $\sin^{-1} x$ is continuous on $[-1, 1]$, and hence on $[0, 1]$.
- (ii) $\frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$, which exists for $-1 < x < 1$. So f is differentiable on $(0, 1)$. (It is not differentiable at $x = 1$, but the theorem requires differentiability only on the *open* interval.)

Both conditions are satisfied.

Next, we compute the slope of the chord. Substituting the end points,

$$f(a) = f(0) = \sin^{-1} 0 = 0, \quad f(b) = f(1) = \sin^{-1} 1 = \frac{\pi}{2},$$

so

$$\frac{f(b) - f(a)}{b - a} = \frac{\frac{\pi}{2} - 0}{1 - 0} = \frac{\pi}{2}.$$

Differentiating f , we get

$$f'(x) = \frac{1}{\sqrt{1-x^2}}, \quad \text{so} \quad f'(c) = \frac{1}{\sqrt{1-c^2}}.$$

By LMVT, there exists a $c \in (0, 1)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$. That is,

$$\frac{1}{\sqrt{1-c^2}} = \frac{\pi}{2}.$$

Taking the reciprocal of both sides, we get

$$\sqrt{1-c^2} = \frac{2}{\pi}.$$

Squaring both sides, we get

$$1 - c^2 = \frac{4}{\pi^2}.$$

Moving c^2 to the right-hand side and $\frac{4}{\pi^2}$ to the left-hand side, we get

$$c^2 = 1 - \frac{4}{\pi^2} = \frac{\pi^2 - 4}{\pi^2}.$$

Taking the positive root, since $c \in (0, 1)$,

$$c = \frac{\sqrt{\pi^2 - 4}}{\pi} = \frac{\sqrt{9.8696 - 4}}{3.1416} = \frac{2.4227}{3.1416} \approx 0.7712,$$

which lies in $(0, 1)$.

Lagrange's mean value theorem is verified, with $c = \frac{\sqrt{\pi^2 - 4}}{\pi} \approx 0.7712$.

Problem 2.3. [Tutorial 1, Q3(c)] Verify Lagrange's mean value theorem for $f(x) = 2x^3 - 3x + 1$ on $[-2, 2]$.

Solution: We are given $f(x) = 2x^3 - 3x + 1$, with $a = -2$ and $b = 2$.

We check the conditions of Lagrange's mean value theorem.

- (i) f is a polynomial, so it is continuous on $[-2, 2]$.

(ii) f is a polynomial, so it is differentiable on $(-2, 2)$.

Both conditions are satisfied.

Next, we compute the slope of the chord. Substituting the end points,

$$f(a) = f(-2) = 2(-2)^3 - 3(-2) + 1 = 2(-8) + 6 + 1 = -16 + 6 + 1 = -9,$$

$$f(b) = f(2) = 2(2)^3 - 3(2) + 1 = 2(8) - 6 + 1 = 16 - 6 + 1 = 11,$$

so

$$\frac{f(b) - f(a)}{b - a} = \frac{11 - (-9)}{2 - (-2)} = \frac{20}{4} = 5.$$

Next, we differentiate f . Using $\frac{d}{dx}(x^n) = nx^{n-1}$,

$$f'(x) = 2(3x^2) - 3 + 0 = 6x^2 - 3, \quad \text{so} \quad f'(c) = 6c^2 - 3.$$

By LMVT, there exists a $c \in (-2, 2)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$. That is,

$$6c^2 - 3 = 5.$$

Moving 3 to the right-hand side, we get $6c^2 = 8$. Dividing both sides by 6, we get

$$c^2 = \frac{8}{6} = \frac{4}{3}.$$

Taking the square root on both sides, we get

$$c = \pm\sqrt{\frac{4}{3}} = \pm\frac{2}{\sqrt{3}} \approx \pm 1.1547.$$

Both values lie in $(-2, 2)$.

Lagrange's mean value theorem is verified, with $c = \pm\frac{2}{\sqrt{3}} \approx \pm 1.1547$.

Problem 2.4. [Tutorial 1, Q4] Apply Lagrange's mean value theorem to $f(x) = \log x$ to prove that

$$0 < \frac{1}{\log(1+x)} - \frac{1}{x} < 1 \quad \text{for } x > 0.$$

Solution: We are given $f(t) = \log t$ on $[1, 1+x]$, where $x > 0$. So $a = 1$ and $b = 1+x$. (The variable of f is written t , since x is already in use.)

We check the conditions of Lagrange's mean value theorem.

(i) $\log t$ is continuous for $t > 0$, so it is continuous on $[1, 1+x]$.

(ii) $\log t$ is differentiable for $t > 0$, so it is differentiable on $(1, 1+x)$.

Both conditions are satisfied.

Next, we compute the slope of the chord. Substituting the end points,

$$f(a) = f(1) = \log 1 = 0, \quad f(b) = f(1+x) = \log(1+x),$$

so

$$\frac{f(b) - f(a)}{b - a} = \frac{\log(1+x) - 0}{(1+x) - 1} = \frac{\log(1+x)}{x}.$$

Next, we differentiate f . Using $\frac{d}{dt}(\log t) = \frac{1}{t}$,

$$f'(t) = \frac{1}{t}, \quad \text{so} \quad f'(c) = \frac{1}{c}.$$

By LMVT, there exists a $c \in (1, 1+x)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$. That is,

$$\frac{1}{c} = \frac{\log(1+x)}{x}.$$

Since $1+x > 1$, we have $\log(1+x) > 0$, so both sides are non-zero. Taking the reciprocal of both sides, we get

$$c = \frac{x}{\log(1+x)}.$$

Since $1 < c < 1+x$,

$$1 < \frac{x}{\log(1+x)} < 1+x.$$

Dividing throughout by $x > 0$, which keeps the inequalities in the same direction,

$$\frac{1}{x} < \frac{1}{\log(1+x)} < \frac{1+x}{x} = \frac{1}{x} + 1.$$

Subtracting $\frac{1}{x}$ throughout,

$$0 < \frac{1}{\log(1+x)} - \frac{1}{x} < 1.$$

$$0 < \frac{1}{\log(1+x)} - \frac{1}{x} < 1 \text{ for every } x > 0, \text{ as required.}$$

Problem 2.5. [Tutorial 1, Q12] Suppose that f is continuous on $[-7, 0]$ and differentiable on $(-7, 0)$, with $f(-7) = -3$ and $f'(x) \leq 2$ for all $x \in (-7, 0)$. Determine the largest possible value of $f(0)$.

Solution: We are given $a = -7$, $b = 0$, $f(a) = f(-7) = -3$, and $f'(x) \leq 2$ on $(-7, 0)$. Also, f is continuous on $[-7, 0]$ and differentiable on $(-7, 0)$.

LMVT states that if f is continuous on $[a, b]$ and differentiable on (a, b) , then there exists a $c \in (a, b)$ such that $f'(c) = \frac{f(b) - f(a)}{b - a}$. Since the function f satisfies the two conditions, applying LMVT to f gives a $c \in (-7, 0)$ such that

$$f'(c) = \frac{f(0) - f(-7)}{0 - (-7)} = \frac{f(0) - (-3)}{7} = \frac{f(0) + 3}{7}.$$

Since $f'(c) \leq 2$, we get

$$\frac{f(0) + 3}{7} \leq 2.$$

Multiplying both sides by 7 (a positive number, so the inequality is unchanged), we get

$$f(0) + 3 \leq 14.$$

Subtracting 3 from both sides, we get

$$f(0) \leq 11.$$

Finally, we show that this bound is attained. Take $f(x) = 2x + 11$. Then f is continuous and differentiable, and

$$f(-7) = 2(-7) + 11 = -14 + 11 = -3, \quad f'(x) = 2 \leq 2, \quad f(0) = 2(0) + 11 = 11.$$

So the value 11 is actually reached.

The largest possible value of $f(0)$ is 11.

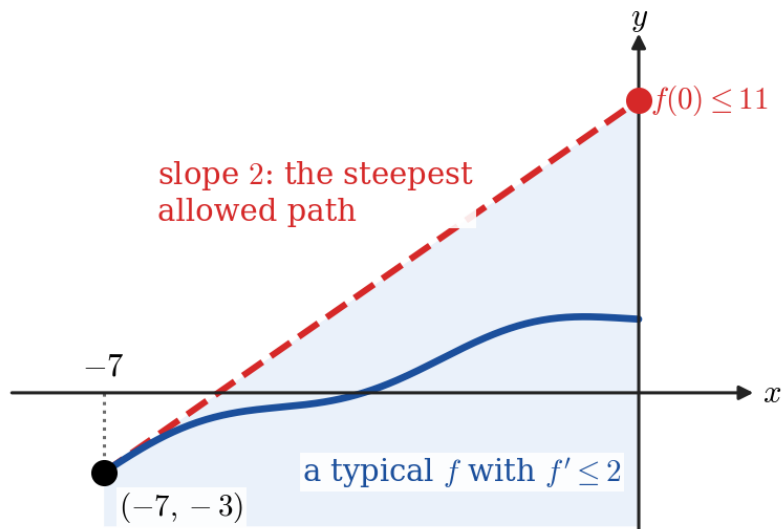


Figure 2.2: Starting from $(-7, -3)$ with slope never exceeding 2, no graph can rise above the line of slope 2. So $f(0) \leq 11$, with equality only for $f(x) = 2x + 11$.

3 Series: the Root and Ratio Tests

Definition 3.1: Convergence of a series

Given a sequence $\{a_n\}$, the series $\sum_{n=1}^{\infty} a_n$ has partial sums $S_n = a_1 + a_2 + \cdots + a_n$. The series *converges* if $\lim_{n \rightarrow \infty} S_n$ exists as a finite number, and *diverges* otherwise.

Theorem 3.2: The n th term test

If $\lim_{n \rightarrow \infty} a_n \neq 0$ (or the limit does not exist), then $\sum a_n$ diverges.

Note:

The converse is false: $a_n \rightarrow 0$ does *not* by itself make $\sum a_n$ converge. The n th term test can only prove divergence.

Theorem 3.3: The root test

Let $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|} = \lim_{n \rightarrow \infty} |a_n|^{1/n}$.

- (i) If $L < 1$, the series $\sum a_n$ converges (absolutely).
- (ii) If $L > 1$ (including $L = \infty$), the series diverges.
- (iii) If $L = 1$, the test gives no information.

Theorem 3.4: The ratio test

Let $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$.

- (i) If $L < 1$, the series $\sum a_n$ converges (absolutely).
- (ii) If $L > 1$ (including $L = \infty$), the series diverges.
- (iii) If $L = 1$, the test gives no information.

Note:

Which test? Use the root test when the general term is an n th power, such as $\left(\frac{2n+3}{3n+2}\right)^n$. Use the ratio test when it contains factorials, or powers such as 2^n and 3^n multiplied by polynomials or logarithms. In both tests, first simplify $|a_n|^{1/n}$ or $\left|\frac{a_{n+1}}{a_n}\right|$ as an expression in n , and only then take the limit.

3.1 Standard limits of sequences

- (i) If $p > 0$, then $\lim_{n \rightarrow \infty} \frac{1}{n^p} = 0$.
- (ii) If $p > 0$, then $\lim_{n \rightarrow \infty} p^{1/n} = 1$.
- (iii) $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$, and more generally $\lim_{n \rightarrow \infty} \left(1 + \frac{k}{n}\right)^n = e^k$ for any constant k .

(iv) $\lim_{n \rightarrow \infty} n^{1/n} = 1.$

(v) If $|x| < 1$, then $\lim_{n \rightarrow \infty} x^n = 0$. In particular $\frac{1}{r^n} = \left(\frac{1}{r}\right)^n \rightarrow 0$ when $r > 1$.

(vi) If $\alpha, \beta > 0$, then $\lim_{n \rightarrow \infty} \frac{n^\alpha}{(1 + \beta)^n} = 0.$

(vii) $\lim_{n \rightarrow \infty} (n!)^{1/n} = \infty$, where $n! = n(n-1)(n-2) \cdots 3 \cdot 2 \cdot 1$.

Note:

In (vi) the numerator n^α is a power function and the denominator $(1 + \beta)^n$ is an exponential function. The exponential grows faster, so the ratio tends to 0.

3.2 Worked problems on the root test

Problem 3.1. [Tutorial 2, Series Q1(a)] Test the convergence of $\sum_{n=1}^{\infty} \left(\frac{2n+3}{3n+2}\right)^n$ by the root test.

Solution: We are given $a_n = \left(\frac{2n+3}{3n+2}\right)^n$. All terms are positive, so $|a_n| = a_n$.

We now compute $|a_n|^{1/n}$. Using $(A^n)^{1/n} = A$,

$$|a_n|^{1/n} = \left[\left(\frac{2n+3}{3n+2}\right)^n\right]^{1/n} = \frac{2n+3}{3n+2}.$$

Next, we take the limit as $n \rightarrow \infty$. Dividing the numerator and the denominator by n ,

$$L = \lim_{n \rightarrow \infty} \frac{2n+3}{3n+2} = \lim_{n \rightarrow \infty} \frac{2 + \frac{3}{n}}{3 + \frac{2}{n}} = \frac{2+0}{3+0} = \frac{2}{3},$$

since $\frac{3}{n} \rightarrow 0$ and $\frac{2}{n} \rightarrow 0$ as $n \rightarrow \infty$.

Since $L = \frac{2}{3} < 1$, the series converges by the root test.

Problem 3.2. [Tutorial 2, Series Q1(b)] Test the convergence of $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$ by the root test.

Solution: We are given $a_n = \left(1 + \frac{1}{n}\right)^{-n^2}$. All terms are positive, so $|a_n| = a_n$.

We now compute $|a_n|^{1/n}$. Using $(A^m)^{1/n} = A^{m/n}$ with $m = -n^2$, so that $\frac{m}{n} = -n$,

$$|a_n|^{1/n} = \left[\left(1 + \frac{1}{n}\right)^{-n^2}\right]^{1/n} = \left(1 + \frac{1}{n}\right)^{-n} = \frac{1}{\left(1 + \frac{1}{n}\right)^n}.$$

Next, we take the limit as $n \rightarrow \infty$. Using the standard limit $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$,

$$L = \lim_{n \rightarrow \infty} \frac{1}{\left(1 + \frac{1}{n}\right)^n} = \frac{1}{e}.$$

Since $L = \frac{1}{e} \approx 0.368 < 1$, the series converges by the root test.

Problem 3.3. [Tutorial 2, Series Q1(h)] Test the convergence of $\sum_{n=1}^{\infty} \left(1 + \frac{2}{n}\right)^{n^2}$ by the root test.

Solution: We are given $a_n = \left(1 + \frac{2}{n}\right)^{n^2}$. All terms are positive, so $|a_n| = a_n$.

We now compute $|a_n|^{1/n}$. Using $(A^m)^{1/n} = A^{m/n}$ with $m = n^2$, so that $\frac{m}{n} = n$,

$$|a_n|^{1/n} = \left[\left(1 + \frac{2}{n}\right)^{n^2} \right]^{1/n} = \left(1 + \frac{2}{n}\right)^n.$$

Next, we take the limit as $n \rightarrow \infty$. Using the standard limit $\lim_{n \rightarrow \infty} \left(1 + \frac{k}{n}\right)^n = e^k$ with $k = 2$,

$$L = \lim_{n \rightarrow \infty} \left(1 + \frac{2}{n}\right)^n = e^2.$$

Since $L = e^2 \approx 7.39 > 1$, the series diverges by the root test.

3.3 Worked problems on the ratio test

Problem 3.4. [Tutorial 2, Series Q2(a)] Test the convergence of $\sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n + 1}$ by the ratio test.

Solution: We are given $a_n = \frac{2^{n-1}}{3^n + 1}$. All terms are positive.

Replacing n by $n + 1$ in a_n ,

$$a_{n+1} = \frac{2^{(n+1)-1}}{3^{n+1} + 1} = \frac{2^n}{3^{n+1} + 1}.$$

We now compute $\left| \frac{a_{n+1}}{a_n} \right|$:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{2^n}{3^{n+1} + 1} \div \frac{2^{n-1}}{3^n + 1} = \frac{2^n}{3^{n+1} + 1} \cdot \frac{3^n + 1}{2^{n-1}} = \frac{2^n}{2^{n-1}} \cdot \frac{3^n + 1}{3^{n+1} + 1} = \frac{2(3^n + 1)}{3^{n+1} + 1},$$

using $\frac{2^n}{2^{n-1}} = 2^{n-(n-1)} = 2$.

Next, we take the limit as $n \rightarrow \infty$. Dividing the numerator and the denominator by 3^n , and using $\frac{3^{n+1}}{3^n} = 3$,

$$L = \lim_{n \rightarrow \infty} \frac{2\left(1 + \frac{1}{3^n}\right)}{3 + \frac{1}{3^n}} = \frac{2(1 + 0)}{3 + 0} = \frac{2}{3},$$

since $\frac{1}{3^n} \rightarrow 0$ as $n \rightarrow \infty$.

Since $L = \frac{2}{3} < 1$, the series converges by the ratio test.

Problem 3.5. [Tutorial 2, Series Q2(d)] Test the convergence of $\sum_{n=0}^{\infty} \frac{n^3 + a}{2^n + a}$, where $a > 0$ is a constant, by the ratio test.

Solution: We are given $a_n = \frac{n^3 + a}{2^n + a}$, with $a > 0$. All terms are positive.

Replacing n by $n + 1$ in a_n ,

$$a_{n+1} = \frac{(n+1)^3 + a}{2^{n+1} + a}.$$

We now compute $\left| \frac{a_{n+1}}{a_n} \right|$:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)^3 + a}{2^{n+1} + a} \cdot \frac{2^n + a}{n^3 + a} = \frac{(n+1)^3 + a}{n^3 + a} \cdot \frac{2^n + a}{2^{n+1} + a}.$$

Dividing the numerator and denominator of the first factor by n^3 , and of the second factor by 2^n ,

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{\left(1 + \frac{1}{n}\right)^3 + \frac{a}{n^3}}{1 + \frac{a}{n^3}} \cdot \frac{1 + \frac{a}{2^n}}{2 + \frac{a}{2^n}},$$

where we used $\frac{(n+1)^3}{n^3} = \left(1 + \frac{1}{n}\right)^3$ and $\frac{2^{n+1}}{2^n} = 2$.

Next, we take the limit as $n \rightarrow \infty$. As $n \rightarrow \infty$: $\frac{1}{n} \rightarrow 0$, $\frac{a}{n^3} \rightarrow 0$ and $\frac{a}{2^n} \rightarrow 0$. Hence

$$L = \frac{(1+0)^3 + 0}{1+0} \cdot \frac{1+0}{2+0} = 1 \cdot \frac{1}{2} = \frac{1}{2}.$$

Since $L = \frac{1}{2} < 1$, the series converges by the ratio test.

Problem 3.6. [Tutorial 2, Series Q2(f)] Test the convergence of $\sum_{n=2}^{\infty} \frac{3^{n+2}}{\log n}$ by the ratio test.

Solution: We are given $a_n = \frac{3^{n+2}}{\log n}$. For $n \geq 2$, $\log n > 0$, so all terms are positive.

Replacing n by $n + 1$ in a_n ,

$$a_{n+1} = \frac{3^{(n+1)+2}}{\log(n+1)} = \frac{3^{n+3}}{\log(n+1)}.$$

We now compute $\left| \frac{a_{n+1}}{a_n} \right|$:

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{3^{n+3}}{\log(n+1)} \cdot \frac{\log n}{3^{n+2}} = \frac{3^{n+3}}{3^{n+2}} \cdot \frac{\log n}{\log(n+1)} = 3 \cdot \frac{\log n}{\log(n+1)}.$$

Using $\log(AB) = \log A + \log B$,

$$\log(n+1) = \log \left[n \left(1 + \frac{1}{n} \right) \right] = \log n + \log \left(1 + \frac{1}{n} \right).$$

Dividing the numerator and the denominator by $\log n$,

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{3 \log n}{\log n + \log \left(1 + \frac{1}{n}\right)} = \frac{3}{1 + \frac{\log \left(1 + \frac{1}{n}\right)}{\log n}}.$$

Next, we take the limit as $n \rightarrow \infty$. As $n \rightarrow \infty$, $\log \left(1 + \frac{1}{n}\right) \rightarrow \log 1 = 0$ while $\log n \rightarrow \infty$, so the fraction $\frac{\log \left(1 + \frac{1}{n}\right)}{\log n} \rightarrow 0$. Hence

$$L = \frac{3}{1 + 0} = 3.$$

Since $L = 3 > 1$, the series diverges by the ratio test.

3.4 A series depending on a variable

When the general term contains a variable x , the test returns a limit L that depends on x . The conditions $L < 1$ and $L > 1$ then become conditions on x , and the values of x with $L = 1$ must be examined separately.

Problem 3.7. [Tutorial 2, Series Q3] Examine the convergence of the series $\sum_{n=1}^{\infty} \left(\frac{nx}{n+1}\right)^n$, where x is real.

Solution: We are given $a_n = \left(\frac{nx}{n+1}\right)^n$, so $|a_n| = \left(\frac{n|x|}{n+1}\right)^n$.

We now compute $|a_n|^{1/n}$. Using $(A^n)^{1/n} = A$,

$$|a_n|^{1/n} = \left[\left(\frac{n|x|}{n+1}\right)^n \right]^{1/n} = \frac{n|x|}{n+1} = \frac{|x|}{1 + \frac{1}{n}},$$

dividing the numerator and the denominator by n .

Next, we take the limit as $n \rightarrow \infty$. Since $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$,

$$L = \lim_{n \rightarrow \infty} \frac{|x|}{1 + \frac{1}{n}} = \frac{|x|}{1 + 0} = |x|.$$

We now consider the cases $|x| < 1$, $|x| > 1$ and $|x| = 1$.

- If $|x| < 1$, then $L < 1$, and the series converges by the root test.
- If $|x| > 1$, then $L > 1$, and the series diverges by the root test.
- If $|x| = 1$, then $L = 1$, and the root test is inconclusive. We use the *n*th term test: if $\lim_{n \rightarrow \infty} a_n \neq 0$, then $\sum a_n$ diverges. For $|x| = 1$,

$$|a_n| = \left(\frac{n}{n+1}\right)^n = \frac{1}{\left(\frac{n+1}{n}\right)^n} = \frac{1}{\left(1 + \frac{1}{n}\right)^n} \rightarrow \frac{1}{e} \neq 0,$$

using the standard limit $\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$. For $x = 1$, the terms tend to $\frac{1}{e}$. For $x = -1$, they alternate in sign, with size approaching $\frac{1}{e}$. In both cases $a_n \not\rightarrow 0$, so the series diverges by the *n*th term test.

The series converges for $-1 < x < 1$ and diverges for $|x| \geq 1$.

4 Taylor and Maclaurin Series

Definition 4.1: Taylor and Maclaurin series

If f has derivatives of all orders at $x = a$, its *Taylor series* about $x = a$ is

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \frac{f'''(a)}{3!}(x-a)^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

The special case $a = 0$ is the *Maclaurin series*

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

Note:

The sum of the terms up to $(x-a)^k$ is the *Taylor polynomial of degree k* . It approximates $f(x)$ well when x is close to a . The equality of f with its full series holds on an interval of convergence, which can be found by the ratio test. Inside that interval a power series may be differentiated or integrated term by term.

Note:

Method. Compute $f(a), f'(a), f''(a), \dots$ as far as needed, divide the n th derivative by $n!$, and attach $(x-a)^n$. Arranging the derivatives in a table keeps the work tidy.

4.1 Worked problems

Problem 4.1. [Tutorial 2, Taylor Q1] Obtain the Maclaurin series expansion, as an infinite series, of each of the following functions.

- (a) $\sin x$.
- (b) $\cos x$.
- (c) e^x .
- (d) $\log(1+x)$.
- (e) $(1+x)^k$, where k is a real constant.

Solution: We use the Maclaurin series. If f has derivatives of all orders at $x = 0$, then

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0) + \frac{x^5}{5!} f^{(5)}(0) + \cdots.$$

- (a) Let $f(x) = \sin x$.

We compute the derivatives of f and their values at $x = 0$. Using $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$:

	f	f'	f''	f'''	$f^{(4)}$	$f^{(5)}$
at x	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$
at $x = 0$	0	1	0	-1	0	1

Substituting these values into the Maclaurin series, we get

$$\sin x = 0 + x(1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(-1) + \frac{x^4}{4!}(0) + \frac{x^5}{5!}(1) + \cdots = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

Observe that the values at 0 repeat in the cycle 0, 1, 0, -1, so only odd powers occur, with alternating signs:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \text{ valid for all real } x.$$

(b) Let $f(x) = \cos x$.

We compute the derivatives of f and their values at $x = 0$. Using $\frac{d}{dx}(\cos x) = -\sin x$ and $\frac{d}{dx}(\sin x) = \cos x$:

	f	f'	f''	f'''	$f^{(4)}$	$f^{(5)}$
at x	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$	$-\sin x$
at $x = 0$	1	0	-1	0	1	0

Substituting these values into the Maclaurin series, we get

$$\cos x = 1 + x(0) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}(1) + \frac{x^5}{5!}(0) + \cdots = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Observe that the values at 0 repeat in the cycle 1, 0, -1, 0, so only even powers occur, with alternating signs:

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \text{ valid for all real } x.$$

(c) Let $f(x) = e^x$.

We compute the derivatives of f and their values at $x = 0$. Using $\frac{d}{dx}(e^x) = e^x$, every derivative is e^x , and $e^0 = 1$:

	f	f'	f''	f'''	$f^{(4)}$
at x	e^x	e^x	e^x	e^x	e^x
at $x = 0$	1	1	1	1	1

Substituting these values into the Maclaurin series, we get

$$e^x = 1 + x(1) + \frac{x^2}{2!}(1) + \frac{x^3}{3!}(1) + \frac{x^4}{4!}(1) + \cdots$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \text{ valid for all real } x.$$

(d) Let $f(x) = \log(1+x)$.

We compute the derivatives of f and their values at $x = 0$. Using $\frac{d}{dx} \log(1+x) = \frac{1}{1+x} = (1+x)^{-1}$ and $\frac{d}{dx}(1+x)^m = m(1+x)^{m-1}$:

$$f'(x) = (1+x)^{-1}, \quad f''(x) = -(1+x)^{-2}, \quad f'''(x) = 2(1+x)^{-3}, \quad f^{(4)}(x) = -6(1+x)^{-4}.$$

	f	f'	f''	f'''	$f^{(4)}$
at x	$\log(1+x)$	$\frac{1}{1+x}$	$-\frac{1}{(1+x)^2}$	$\frac{2}{(1+x)^3}$	$-\frac{6}{(1+x)^4}$
at $x = 0$	0	1	-1	2	-6

Substituting these values into the Maclaurin series, we get

$$\begin{aligned} \log(1+x) &= 0 + x(1) + \frac{x^2}{2!}(-1) + \frac{x^3}{3!}(2) + \frac{x^4}{4!}(-6) + \dots \\ &= x - \frac{x^2}{2} + \frac{2x^3}{6} - \frac{6x^4}{24} + \dots = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots \end{aligned}$$

Observe that the term in x^n is $\frac{x^n}{n}$, and that the signs alternate. Therefore,

$$\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots = \sum_{n=1}^{\infty} \frac{(-1)^{n-1} x^n}{n}, \quad \text{valid for } -1 < x \leq 1.$$

(e) Let $f(x) = (1+x)^k$, with k a real constant.

We compute the derivatives of f and their values at $x = 0$. Using $\frac{d}{dx}(1+x)^m = m(1+x)^{m-1}$ repeatedly:

	at x	at $x = 0$
f	$(1+x)^k$	1
f'	$k(1+x)^{k-1}$	k
f''	$k(k-1)(1+x)^{k-2}$	$k(k-1)$
f'''	$k(k-1)(k-2)(1+x)^{k-3}$	$k(k-1)(k-2)$
$f^{(4)}$	$k(k-1)(k-2)(k-3)(1+x)^{k-4}$	$k(k-1)(k-2)(k-3)$

Substituting these values into the Maclaurin series, we get

$$(1+x)^k = 1 + x(k) + \frac{x^2}{2!} k(k-1) + \frac{x^3}{3!} k(k-1)(k-2) + \frac{x^4}{4!} k(k-1)(k-2)(k-3) + \dots$$

$$(1+x)^k = 1 + kx + \frac{k(k-1)}{2!}x^2 + \frac{k(k-1)(k-2)}{3!}x^3 + \frac{k(k-1)(k-2)(k-3)}{4!}x^4 + \dots$$

(the binomial series), valid for $|x| < 1$. When k is a non-negative integer, the series stops at x^k and holds for all x .

Problem 4.2. [Tutorial 2, Taylor Q2] Obtain the Taylor series expansion of $\log_e x$ about $x = 1$ up to the term containing $(x - 1)^4$, and hence find the approximate value of $\log_e(1.1)$.

Solution: We are given $f(x) = \log x$, with $a = 1$.

We use the Taylor series of f about $x = a$:

$$f(x) = f(a) + (x - a)f'(a) + \frac{(x - a)^2}{2!}f''(a) + \frac{(x - a)^3}{3!}f'''(a) + \frac{(x - a)^4}{4!}f^{(4)}(a) + \dots$$

We compute the derivatives of f and their values at $x = 1$. Using $\frac{d}{dx}(\log x) = \frac{1}{x} = x^{-1}$ and $\frac{d}{dx}(x^m) = mx^{m-1}$:

$$f'(x) = x^{-1}, \quad f''(x) = -x^{-2}, \quad f'''(x) = 2x^{-3}, \quad f^{(4)}(x) = -6x^{-4}.$$

	f	f'	f''	f'''	$f^{(4)}$
at x	$\log x$	$\frac{1}{x}$	$-\frac{1}{x^2}$	$\frac{2}{x^3}$	$-\frac{6}{x^4}$
at $x = 1$	$\log 1 = 0$	1	-1	2	-6

Substituting these values into the Taylor series with $a = 1$, we get

$$\begin{aligned} \log x &= 0 + (x - 1)(1) + \frac{(x - 1)^2}{2!}(-1) + \frac{(x - 1)^3}{3!}(2) + \frac{(x - 1)^4}{4!}(-6) + \dots \\ &= (x - 1) - \frac{(x - 1)^2}{2} + \frac{2(x - 1)^3}{6} - \frac{6(x - 1)^4}{24} + \dots \\ &= (x - 1) - \frac{(x - 1)^2}{2} + \frac{(x - 1)^3}{3} - \frac{(x - 1)^4}{4} + \dots \end{aligned}$$

To approximate $\log(1.1)$, we put $x = 1.1$, so that $x - 1 = 0.1$:

$$\begin{aligned} \log(1.1) &\approx 0.1 - \frac{(0.1)^2}{2} + \frac{(0.1)^3}{3} - \frac{(0.1)^4}{4} = 0.1 - \frac{0.01}{2} + \frac{0.001}{3} - \frac{0.0001}{4} \\ &= 0.1 - 0.005 + 0.000333 - 0.000025 = 0.095308. \end{aligned}$$

$$\log x = (x - 1) - \frac{(x - 1)^2}{2} + \frac{(x - 1)^3}{3} - \frac{(x - 1)^4}{4} + \dots, \text{ and } \log(1.1) \approx 0.0953.$$

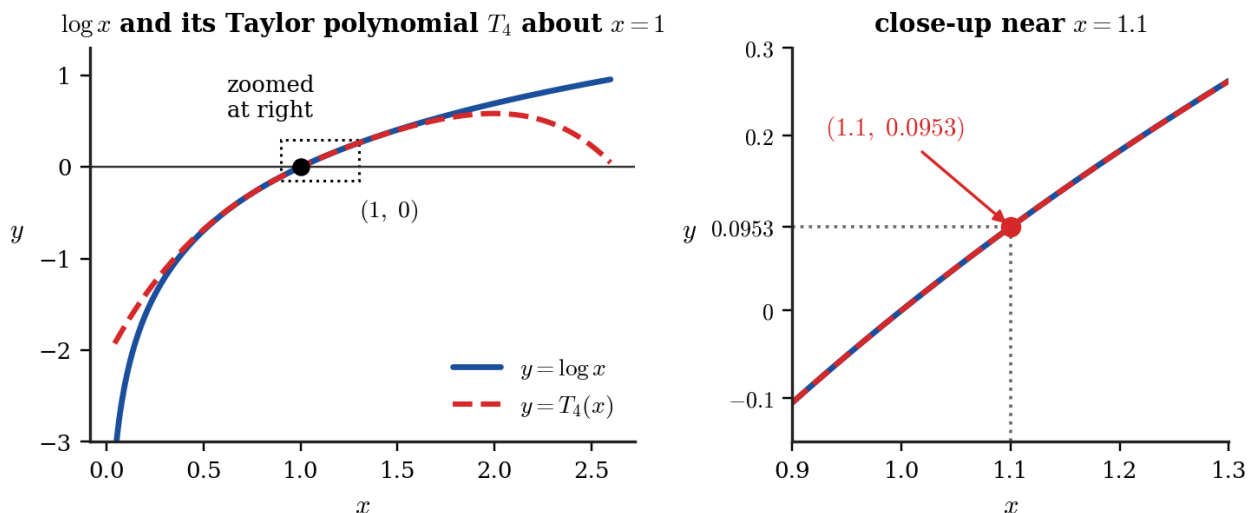


Figure 4.1: Left: $\log x$ and its Taylor polynomial $T_4(x)$ of degree 4 about $x = 1$, which pass through (1, 0). Right: a close-up of the dotted box. At $x = 1.1$ the polynomial gives $T_4(1.1) = 0.0953$, and the two graphs cannot be told apart (the true value is $\log 1.1 = 0.0953102\dots$).

Problem 4.3. [Tutorial 2, Taylor Q4] Obtain the Taylor series expansion of $f(x) = \sin x$ about $x = \frac{\pi}{6}$ up to the term containing $(x - \frac{\pi}{6})^5$.

Solution: We are given $f(x) = \sin x$, with $a = \frac{\pi}{6}$, with $\sin \frac{\pi}{6} = \frac{1}{2}$ and $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$.

We use the Taylor series of f about $x = a$:

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \frac{(x-a)^4}{4!}f^{(4)}(a) + \frac{(x-a)^5}{5!}f^{(5)}(a) + \dots$$

We compute the derivatives of f and their values at $x = \frac{\pi}{6}$. Using $\frac{d}{dx}(\sin x) = \cos x$ and $\frac{d}{dx}(\cos x) = -\sin x$:

	f	f'	f''	f'''	$f^{(4)}$	$f^{(5)}$
at x	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$
at $x = \frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$

Substituting these values into the Taylor series, and writing $h = x - \frac{\pi}{6}$ for short, we get

$$\begin{aligned} \sin x &= \frac{1}{2} + h \left(\frac{\sqrt{3}}{2} \right) + \frac{h^2}{2!} \left(-\frac{1}{2} \right) + \frac{h^3}{3!} \left(-\frac{\sqrt{3}}{2} \right) + \frac{h^4}{4!} \left(\frac{1}{2} \right) + \frac{h^5}{5!} \left(\frac{\sqrt{3}}{2} \right) + \dots \\ &= \frac{1}{2} + \frac{\sqrt{3}}{2}h - \frac{1}{2 \cdot 2}h^2 - \frac{\sqrt{3}}{6 \cdot 2}h^3 + \frac{1}{24 \cdot 2}h^4 + \frac{\sqrt{3}}{120 \cdot 2}h^5 + \dots \\ &= \frac{1}{2} + \frac{\sqrt{3}}{2}h - \frac{1}{4}h^2 - \frac{\sqrt{3}}{12}h^3 + \frac{1}{48}h^4 + \frac{\sqrt{3}}{240}h^5 + \dots \end{aligned}$$

$$\sin x = \frac{1}{2} + \frac{\sqrt{3}}{2} \left(x - \frac{\pi}{6} \right) - \frac{1}{4} \left(x - \frac{\pi}{6} \right)^2 - \frac{\sqrt{3}}{12} \left(x - \frac{\pi}{6} \right)^3 + \frac{1}{48} \left(x - \frac{\pi}{6} \right)^4 + \frac{\sqrt{3}}{240} \left(x - \frac{\pi}{6} \right)^5 + \dots$$

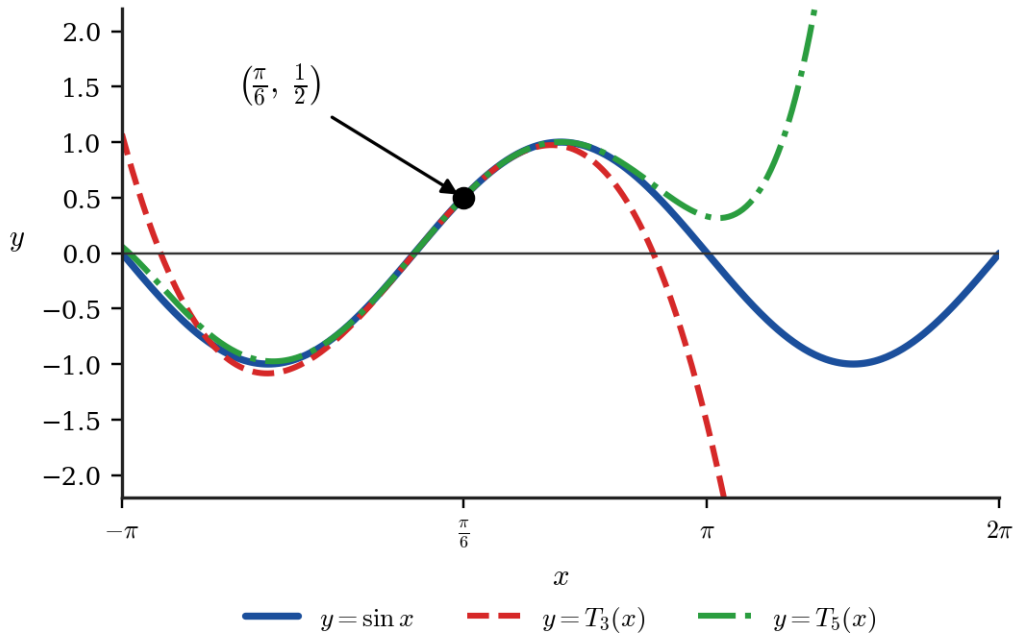


Figure 4.2: $\sin x$ and its Taylor polynomials T_3 and T_5 of degrees 3 and 5 about $a = \frac{\pi}{6}$. All three pass through the point $(\frac{\pi}{6}, \frac{1}{2})$, and the degree-5 polynomial stays close to $\sin x$ over a wider interval.

Problem 4.4. [Tutorial 2, Taylor Q11] Evaluate $\int e^{-x^2} dx$ as an infinite series.

Solution: We are given the integral $\int e^{-x^2} dx$. It cannot be evaluated in terms of elementary functions, so we integrate its Maclaurin series.

We first compute the Maclaurin series of e^x . The Maclaurin series of a function f is

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{(4)}(0) + \cdots$$

For $f(x) = e^x$, every derivative is e^x , so $f(0) = f'(0) = f''(0) = \cdots = e^0 = 1$. Hence

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \cdots, \quad \text{valid for all real } x.$$

Replacing x with $-x^2$ in the above expression, we get

$$e^{-x^2} = 1 + (-x^2) + \frac{(-x^2)^2}{2!} + \frac{(-x^2)^3}{3!} + \frac{(-x^2)^4}{4!} + \cdots$$

Since $(-x^2)^n = (-1)^n x^{2n}$, this simplifies to

$$e^{-x^2} = 1 - x^2 + \frac{x^4}{2!} - \frac{x^6}{3!} + \frac{x^8}{4!} - \cdots = 1 - x^2 + \frac{x^4}{2} - \frac{x^6}{6} + \frac{x^8}{24} - \cdots,$$

which is valid for all real x .

Integrating the above equation term by term, which is allowed for a power series, we obtain

$$\int e^{-x^2} dx = \int 1 dx - \int x^2 dx + \frac{1}{2} \int x^4 dx - \frac{1}{6} \int x^6 dx + \cdots$$

Using $\int x^m dx = \frac{x^{m+1}}{m+1}$ for each term,

$$\int e^{-x^2} dx = x - \frac{x^3}{3} + \frac{1}{2} \cdot \frac{x^5}{5} - \frac{1}{6} \cdot \frac{x^7}{7} + \cdots + C = x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \cdots + C,$$

where C is the constant of integration.

$$\int e^{-x^2} dx = x - \frac{x^3}{3} + \frac{x^5}{10} - \frac{x^7}{42} + \cdots + C.$$

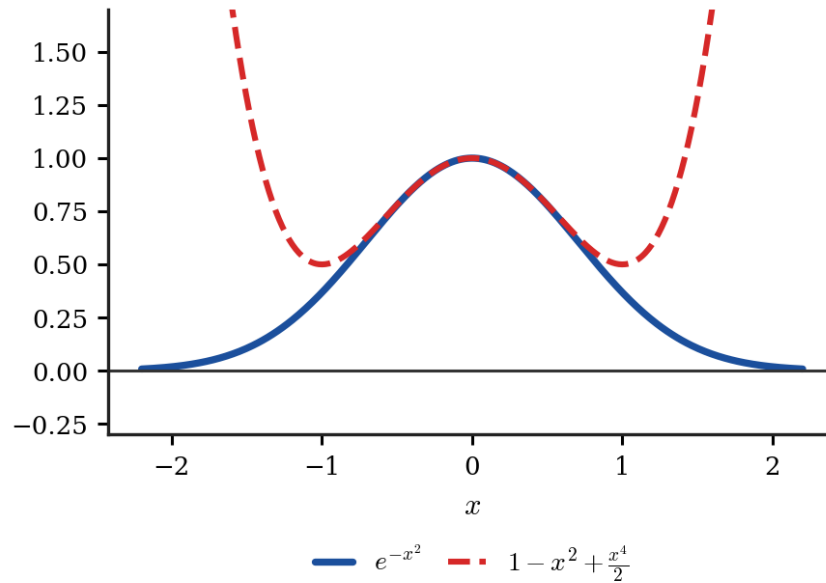


Figure 4.3: e^{-x^2} and the polynomial $1 - x^2 + \frac{x^4}{2}$ formed from its first three terms. They agree closely for $|x| < 0.6$ or so.

References

- [1] J. Stewart, D. K. Clegg and S. Watson, *Calculus: Early Transcendentals*, 9th edition, Cengage Learning, Boston, 2021.